

CBSE Class 10 Mathematics - Answer Key

1. For the system of linear equations $3x + 2y = 8$ and $ax - y = 3$, find the value of a such that the system has a unique solution using the property of coefficients.

Answer: (C)

Solution:

1. Identify the coefficients $a_1 = 3, b_1 = 2, c_1 = 8$ and $a_2 = a, b_2 = -1, c_2 = 3$.
2. For a unique solution, the condition is $a_1/a_2 \neq b_1/b_2$.
3. Substitute the values: $3/a \neq 2/(-1)$.
4. Solve the inequality: $3/a \neq -2 \Rightarrow -2a \neq 3 \Rightarrow a \neq -3/2$.

Derivation:

$$\begin{aligned}a_1/a_2 &\neq b_1/b_2 \\3/a &\neq 2/(-1) \\-2a &\neq 3 \\a &\neq -1.5\end{aligned}$$

Pair of Linear Equations in Two Variables — Consistency of Equations

2. The pair of equations $x + 2y = 5$ and $3x + ky + 15 = 0$ represents parallel lines. What is the value of k ?

Answer: (B)

Solution:

1. The condition for parallel lines is $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.
2. Here, $a_1 = 1, b_1 = 2, c_1 = -5$ and $a_2 = 3, b_2 = k, c_2 = 15$.
3. Set the ratio $a_1/a_2 = b_1/b_2$: $1/3 = 2/k$.
4. Solve for k : $k = 6$.

Derivation:

$$\begin{aligned}1/3 &= 2/k \\k &= 3 \times 2 \\k &= 6\end{aligned}$$

Pair of Linear Equations in Two Variables — Graphical Representation

3. If the pair of equations $2x + 3y = 7$ and $(a + b)x + (2a - b)y = 21$ has infinitely many solutions, then which of the following is true?

Answer: (A)

Solution:

1. For infinitely many solutions, $a_1/a_2 = b_1/b_2 = c_1/c_2$.
2. $2/(a+b) = 3/(2a-b) = 7/21 = 1/3$.
3. From $2/(a+b) = 1/3$, $a+b = 6$.
4. From $3/(2a-b) = 1/3$, $2a-b = 9$.
5. Adding equations: $3a = 15 \Rightarrow a = 5, b = 1$.

Derivation:

$$\begin{aligned}a + b &= 6 \\2a - b &= 9 \\3a &= 15 \\a = 5, b &= 1\end{aligned}$$

Pair of Linear Equations in Two Variables — Consistency of Equations

4. For what value of k does the system $kx + 2y = 5$ and $3x + y = 1$ have a unique solution?

Answer: (D)

Solution:

1. For a unique solution, $a_1/a_2 \neq b_1/b_2$.
2. $k/3 \neq 2/1$.
3. $k \neq 6$.

Derivation:

$$\begin{aligned}k/3 &\neq 2/1 \\k &\neq 6\end{aligned}$$

Pair of Linear Equations in Two Variables — Consistency of Equations

5. Consider the system $4x - 3y = 9$ and $2x + ky = 11$. If the system is consistent and independent, what must be true about k ?

Answer: (C)

Solution:

1. Consistent and independent means a unique solution exists.
2. Condition: $a_1/a_2 \neq b_1/b_2$.
3. $4/2 \neq -3/k$.
4. $2 \neq -3/k \Rightarrow 2k \neq -3 \Rightarrow k \neq -1.5$.

Derivation:

$$\begin{aligned}4/2 &\neq -3/k \\2 &\neq -3/k \\k &\neq -3/2\end{aligned}$$

6. Which of the following conditions represents a system of linear equations that has no solution?

Answer: (B)

Solution:

1. A system has no solution if the lines are parallel.
2. The condition for parallel lines is $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.

Derivation:

$$a_1/a_2 = b_1/b_2 \neq c_1/c_2$$

7. If the pair of equations $x - y = 1$ and $2x + ky = 2$ is coincident, find k .

Answer: (C)

Solution:

1. Coincident lines satisfy $a_1/a_2 = b_1/b_2 = c_1/c_2$.
2. $1/2 = -1/k = -1/-2$.
3. $1/2 = -1/k \Rightarrow k = -2$.

Derivation:

$$\begin{aligned} 1/2 &= -1/k \\ k &= -2 \end{aligned}$$

8. The pair of equations $\frac{2}{x} + \frac{3}{y} = 13$ and $\frac{5}{x} - \frac{4}{y} = -2$ can be reduced to a linear system in u and v by substituting $u = \frac{1}{x}$ and $v = \frac{1}{y}$. If the resulting system is $2u + 3v = 13$ and $5u - 4v = -2$, what is the value of $x + y$?

Answer: (A)

Solution:

1. Substitute $u = 1/x$ and $v = 1/y$ to get the linear system: $2u + 3v = 13$ (eq 1) and $5u - 4v = -2$ (eq 2).
2. Multiply (eq 1) by 4 and (eq 2) by 3: $8u + 12v = 52$ and $15u - 12v = -6$.
3. Add the two equations: $23u = 46$, which gives $u = 2$.
4. Substitute $u = 2$ into (eq 1): $2(2) + 3v = 13 \Rightarrow 3v = 9 \Rightarrow v = 3$.
5. Find x and y : $x = 1/u = 1/2$ and $y = 1/v = 1/3$.
6. Calculate $x + y = 1/2 + 1/3 = 3/6 + 2/6 = 5/6$.

Derivation:

$$\begin{aligned}
2u + 3v &= 13 \quad (1) \\
5u - 4v &= -2 \quad (2) \\
(2u + 3v) \times 4 & \\
\Rightarrow 8u + 12v &= 52 \\
(5u - 4v) \times 3 & \\
\Rightarrow 15u - 12v &= -6 \\
23u &= 46 \\
\Rightarrow u &= 2 \\
2(2) + 3v &= 13 \\
\Rightarrow 3v &= 9 \\
\Rightarrow v &= 3 \\
x &= \frac{1}{2}, y = \frac{1}{3} \\
x + y &= \frac{1}{2} + \frac{1}{3} = \frac{5}{6}
\end{aligned}$$

Pair of Linear Equations in Two Variables — equations reducible to linear form

9. For what value of p is the system $px + 3y = 7$ and $2x + 6y = 14$ inconsistent?

Answer: (D)

Solution:

1. Inconsistent means no solution, so $a_1/a_2 = b_1/b_2 \neq c_1/c_2$.
2. $p/2 = 3/6 \neq 7/14$.
3. $p/2 = 1/2 \Rightarrow p = 1$.
4. Check the third ratio: $1/2 \neq 1/2$ is false (they are equal).
5. Since $1/2 = 1/2 = 1/2$, the lines are coincident (infinitely many solutions) for $p = 1$. Therefore, no value of p makes it inconsistent.

Derivation:

$$\begin{aligned}
p/2 &= 3/6 = 7/14 \\
p/2 &= 1/2 \\
p &= 1
\end{aligned}$$

But $1/2 = 1/2$ means infinite solutions. No p exists.

Pair of Linear Equations in Two Variables — Consistency of Equations

10. If the lines $ax + by = c$ and $dx + ey = f$ are parallel, which relation is correct?

Answer: (B)

Solution:

1. Parallel lines have no solution.
2. Condition: $a/d = b/e \neq c/f$.

Derivation:

$$a/d = b/e \neq c/f$$

Pair of Linear Equations in Two Variables — Consistency of Equations